

Electronic Structure Bench

Where an electron actually is, and how measured ionization energies hand you the shell structure without anyone having to assert it.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/electronic-structure-bench/

1 Before you open anything

Answer from what you already know. The value of this page is having something on paper to argue with later.

PREDICTION

A magnesium atom has 12 electrons. Pull them off one at a time and measure the energy each removal costs. At some point the next electron will suddenly cost several times more than the one before it. **After how many removals does that first jump happen?** Give a whole number, and a rough guess at the factor ($\times 2?$ $\times 5?$ $\times 20?$).

Write one sentence saying what you reasoned from.

2 Getting your bearings: where the electron is

Open the demonstration and stay on tab 1, **Where is the electron?** The left canvas is a slice through the nucleus, one dot per sampled position; the right canvas is the radial distribution. Four readouts sit under **Numbers for this orbital**.

1. Tick **Mark the radial nodes** and **Overlay the Bohr orbit**. Leave **Same zoom for every orbital** unticked for now.
2. Press each orbital button in turn: **1s**, **2s**, **2p**, **3s**, **3p**, **3d z²**.
3. For each one, copy the four readouts into the table. The **Nodes (r / ang)** box gives two numbers separated by a slash: radial nodes first, angular second.

Orbital	Most likely (pm)	90% within (pm)	Bohr's answer (pm)	Radial nodes	Angular nodes
1s					
2s					
2p					
3s					
3p					
3d z^2					

a. In three rows the *Most likely* distance matches *Bohr's answer* exactly. Which three? Write down n and l for each, and state what those three have in common.

b. Using the last two columns, write a formula for the number of radial nodes in terms of n and l . Check it against all six rows.

c. Select **3s** and press **Draw the cloud again** three or four times. Name one thing that changes and one thing that does not, and say what that distinction tells you about what the dots represent.

d. Tick **Same zoom for every orbital** and step from 1s to 3s. Describe what the fixed scale reveals that the automatic scale hid.

Every radius in this table is for a **hydrogen** atom — the exact one-electron wavefunctions. A real sodium 3s orbital has the same shape and the same node count, but is pulled far closer to the nucleus by the larger nuclear charge. Do not quote 375.6 pm as a sodium radius.

3 The core relationship: where the cliff falls

Go to tab 2, **Why ionization energy jumps**. These bars are measured laboratory values, on a logarithmic axis — one gridline is a factor of ten.

1. In **Choose an element**, click **Na**.
2. Drag **First big jump comes after removing** to the number you want to commit to, then press **Reveal the measurements**.
3. Record **Group / period**, **First cliff at** and **Cost multiplier** in the table.
4. Repeat for **Mg**, **Al**, **Si** and **P**. Set the slider each time before revealing.

Touching the slider after you reveal hides the chart again, on purpose. Set your number first, then reveal. Switching elements also re-hides.

Element	Group / period	Your prediction	First cliff at (e^-)	Cost multiplier
Na				
Mg				
Al				
Si				
P				

a. State the rule connecting the *First cliff at* number to the group number, in your own words. Your rule has to work for group 1 and 2 as well as for groups 13–15.

b. The *Cost multiplier* shrinks steadily from Na to P. Nothing about the shell boundary is getting weaker. Suggest what else is changing that makes the *ratio* smaller.

c. Compare the Mg row with your prediction on page 1. If the number or the factor was off, say precisely which assumption produced it.

d. Now click **Ca**, set the slider to **2**, reveal, and read the **Read the fine print** panel. Record the largest single multiplier in the whole chart, which electron it belongs to, and how many occupied shells calcium has.

The first cliff is therefore not the tallest one. Explain in one or two sentences why the *first* cliff is still the one that gives you the valence count.

4 Where the quick answer fails

"First ionization energy increases across a period" is the standard one-line trend, and it is mostly right. Tab 3, **The sawtooth**, shows exactly where it is not. Use the **Straight through, H to Kr** view with **Colour every step that goes down** ticked.

1. Select each element below by tapping its point on the chart, or with **Next** → and ← **Previous**.
2. Record **Group**, **First IE** and **Step from prev**. The step is negative for every one of these.
3. For the last column you need the previous element's value: it is (First IE) – (Step from prev), or press ← **Previous** once and read it.

Element	Group	First IE (kJ/mol)	Step from prev (kJ/mol)	Step as % of the previous element's IE
B				
O				
Al				
S				
Ga				
Se				

a. Only two group numbers appear in that table. Using the **Selected element** note and the configuration shown beneath it, give the structural reason for each of the two drops. They are not the same reason.

b. Tick **Shade groups 13 and 16**. Does every shaded column sit at a drop, in all three periods shown? Quote one value that makes the drop hard to see by eye.

c. An exam asks: "Does first ionization energy increase across period 4?" Write an answer of two or three sentences that is both correct and honest about the exceptions, using your Ga and Se numbers.

d. Switch to **Rows stacked on each other**. Periods 2 and 3 are plotted against group number. Describe what lines up between them, and what the vertical offset between the two curves tells you.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it and check yourself — and where you were wrong, write down which step went off.

1. An unknown element gives these successive ionization energies in kJ/mol: 786.5, 1 577.1, 3 231.6, 4 355.5, 16 091, 19 805. Work out the ratio of each value to the one before it. State after which removal the first cliff falls, how many valence electrons the atom has, and which group it is in. Given that it is in period 3, name it.

2. Write the full electron configuration of calcium. From the configuration alone, predict (i) after how many removals the first cliff appears and (ii) how many large multipliers appear in the complete chart of all twenty ionization energies. Justify each answer from the configuration, not from memory of the chart.

3. Using your node formula from part 2: which orbital in the tab-1 picker has exactly one radial node and exactly one angular node? Then, for a 4d orbital, give the number of radial nodes and the number of angular nodes, and the radius the Bohr model would assign to $n = 4$ (take $a_0 = 52.9 \text{ pm}$).

4. Boron has one more proton than beryllium, and sulfur has one more proton than phosphorus. Both have a *lower* first ionization energy than the element before them. Explain each drop. Your two explanations must differ from each other, and both must be consistent with increasing nuclear charge.

6 On your own

Nothing here is graded. The demonstration holds more than this worksheet used.

- In tab 1, compare **3d xz** with **3d z²**. The clouds look nothing alike and the graph on the right is identical. Work out why, then predict whether 2p and 2s would do the same.
- In tab 2, hunt for the element between Li and Ca with the largest first-cliff *Cost multiplier*. Before you check, argue from configurations which one it ought to be.
- In tab 3, step through Sc to Zn and read the note each time. First ionization energy barely moves across the whole d-block. Say what that flatness implies about which sub-shell the

outermost electron is sitting in.

- Use **Hide again** in tab 2 to quiz someone else: name an element, have them commit to a cliff position, then reveal.